- SGm means SGMII @ 1.25 Gbaud where "m" indicates which MAC on the Frame Manager. For example, "SG3" indicates SGMII for MAC 3 at 1.25 Gbaud.
- *SGm* means SGMII @3.125 Gbaud where "*m*" indicates which MAC on the Frame Manager. For example, "*SG3*" indicates SGMII for MAC 3 at 3.25 Gbaud.

- PCIe notation :
  - PCIem is PCIe @ 5/2.5 GT/s, m indicates the PCIe controller number.
  - *PCIem* is PCIe @ 8/5/2.5 GT/s, *m* indicates the PCIe controller number.
- SRIO notation :
  - SRIOm, m is SRIO1 or SRIO2 @ 5/2.5 Gbaud *SRIOm*, *m* is SRIO1 or SRIO2 @ 3.125 Gbaud
- Aurora notation : Aurora is Aurora @ 5/2.5 Gbps
- SATA notation : SATAm, where m is SATA1 or SATA2 @ 3/1.5 Gbps
- Per lane PLL mapping: 1-PLL1, 2-PLL2

NOTE

The table below lists the SerDes protocol SRDS_PRTCL_S1 and SRDS_PRTCL_S2 combinations supported.When selecting protocols from the table below, there is a restriction to choose the supported protocol combinations by line (SerDes 1 and SerDes 2 together), for example SRDS_PRCTL_S1= A2 can ONLY be combined with SRDS_PRTCL_S2= 16

Table 19-1. SerDes Lanes Assignments and Multiplexing

SERDES1										SERDES2									
SRDS_PRTCL_S1	A	B	C	D	E	F	G	H	Per Lane PLL Mapping	SRDS_PRTCL_S2	A	B	C	D	E	F	G	H	Per Lane PLL Mapping
1C	SG9	SG10	SG1	SG2	SG3	SG4	SG5	SG6	1111 1111	1F[1]	*PCIe1*				PCIe2				1111 2222
95	SG9	SG10	*SG1*	*SG2*	SG3	SG4	SG5	SG6	1122 1111	1F[1]	*PCIe1*				PCIe2				1111 2222
A2	SG9	SG10	SG1	SG2	SG3	SG4	SG5	SG6	1111 1111	16	PCIe1				PCIe2		SATA1	SATA2	1111 1122
94	SG9	SG10	*SG1*	*SG2*	SG3	SG4	SG5	SG6	1122 1111	16	PCIe1				PCIe2		SATA1	SATA2	1111 1122
51	XAUI9				PCIe4	SG4	SG5	SG6	1111	1F[1]	*PCIe1*				PCIe2				1111

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Table 19-1. SerDes Lanes Assignments and Multiplexing (continued)

SERDES1										SERDES2									
SRDS_PRTCL_S1	A	B	C	D	E	F	G	H	Per Lane PLL Mapping	SRDS_PRTCL_S2	A	B	C	D	E	F	G	H	Per Lane PLL Mapping
									2222										2222
5F	Higig9				PCIe4	SG4	SG5	SG6	1111 2222	1F[1]	*PCIe1*				PCIe2				1111 2222
65	*Higig9*				PCIe4	SG4	SG5	SG6	1111 2222	1F[1]	*PCIe1*				PCIe2				1111 2222
6B	XFI9	XFI10	XFI1	XFI2	PCIe4	SG4	SG5	SG6	1111 2222	1F[1]	*PCIe1*				PCIe2				1111 2222
6C	XFI9	XFI10	SG1	SG2	PCIe4				1122 2222	16	PCIe1				PCIe2		SATA1	SATA2	1111 1122
6D	XFI9	XFI10	SG1	SG2	PCIe4				1122 2222	1F[1]	*PCIe1*				PCIe2				1111 2222
6D	XFI9	XFI10	SG1	SG2	PCIe4				1122 2222	1	PCIe1								1111 1111
6C	XFI9	XFI10	SG1	SG2	PCIe4				1122 2222	29	*SRIO2*				*SRIO1*				1111 1111
6D	XFI9	XFI10	SG1	SG2	PCIe4				1122 2222	2D	SRIO2				SRIO1				1111 1111
71	XFI9	XFI10	SG1	SG2	PCIe4	SG4	SG5	SG6	1122 2222	1F[1]	*PCIe1*				PCIe2				1111 2222
71	XFI9	XFI10	SG1	SG2	PCIe4	SG4	SG5	SG6	1122 2222	15	PCIe1				PCIe2		SATA1	SATA2	1111 1122
6C	XFI9	XFI10	SG1	SG2	PCIe4				1122 2222	18	PCIe1				Aurora		SATA1	SATA2	1111 1122
A6[2]	SG9	SG10	SG1	SG2	*PCIe4*		SG5	SG6	1111 2211	29	*SRIO2*				*SRIO1*				1111 1111
A6[2]	SG9	SG10	SG1	SG2	*PCIe4*		SG5	SG6	1111 2211	2E	SRIO2				SRIO1				1111 1111
8E	SG9	SG10	*SG1*	*SG2*	PCIe4		SG5	SG6	1122 1111	29	*SRIO2*				*SRIO1*				1111 1111
8F	SG9	SG10	*SG1*	*SG2*	PCIe4		SG5	SG6	1122 1111	2D	SRIO2				SRIO1				1111 1111
82	*SG9*	*SG10*	*SG1*	*SG2*	PCIe4		SG5	SG6	1111 2222	29	*SRIO2*				*SRIO1*				1111 1111

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Table 19-1. SerDes Lanes Assignments and Multiplexing (continued)

SERDES1										SERDES2									
SRDS_PRTCL_S1	A	B	C	D	E	F	G	H	Per Lane PLL Mapping	SRDS_PRTCL_S2	A	B	C	D	E	F	G	H	Per Lane PLL Mapping
83	*SG9*	*SG10*	*SG1*	*SG2*	PCIe4		SG5	SG6	1111 2222	2D	SRIO2				SRIO1				1111 1111
A4[3]	SG9	SG10	SG1	SG2	*PCIe4*				1111 1111	16	PCIe1				PCIe2		SATA1	SATA2	1111 1122
96	SG9	SG10	*SG1*	SG2	PCIe4				1121 1111	16	PCIe1				PCIe2		SATA1	SATA2	1111 1122
8A	SG9	SG10	*SG1*	*SG2*	PCIe4				1122 1111	16	PCIe1				PCIe2		SATA1	SATA2	1111 1122
AB	PCIe3				PCIe4				1111 1111	1F[1]	*PCIe1*				PCIe2				1111 2222
DA	PCIe3								1111 1111	02	PCIe1								1111 1111
D9	PCIe3	SG10	SG1	SG2	PCIe4	SG4	SG5	SG6	1111 2111	1F[1]	*PCIe1*				PCIe2				1111 2222
D3	PCIe3	SG10	*SG1*	SG2	PCIe4	SG4	SG5	SG6	1121 1111	1F[1]	*PCIe1*				PCIe2				1111 2222
CB	PCIe3	SG10	*SG1*	*SG2*	PCIe4	SG4	SG5	SG6	1122 1111	1F[1]	*PCIe1*				PCIe2				1111 2222
D8[2]	PCIe3	SG10	SG1	SG2	*PCIe4*	SG4	SG5	SG6	1111 2111	16	PCIe1				PCIe2		SATA1	SATA2	1111 1122
6E	XFI9	XFI10	SG1	SG2	PCIe4		SG5	SG6	1122 2222	18	PCIe1				Aurora		SATA1	SATA2	1111 1122
BC[3]	PCIe3		SG1	SG2	*PCIe4*				1111 1111	18	PCIe1				Aurora		SATA1	SATA2	1111 1122
BC[3]	PCIe3		SG1	SG2	*PCIe4*				1111 1111	15	PCIe1				PCIe2		SATA1	SATA2	1111 1122
C8	PCIe3	SG10	*SG1*	*SG2*	PCIe4		SG5	SG6	1122 1111	18	PCIe1				Aurora		SATA1	SATA2	1111 1122
D6[2]	PCIe3	SG10	SG1	SG2	*PCIe4*		SG5	SG6	1111 2211	18	PCIe1				Aurora		SATA1	SATA2	1111 1122
DE [3]	PCIe3				*PCIe4*	PCIe1	PCIe2	SG6	1111 1111	36	SRIO2				Aurora		SATA1	SATA2	1111 1122

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Table 19-1. SerDes Lanes Assignments and Multiplexing (continued)

SERDES1										SERDES2									
SRDS_PRTCL_S1	A	B	C	D	E	F	G	H	Per Lane PLL Mapping	SRDS_PRTCL_S2	A	B	C	D	E	F	G	H	Per Lane PLL Mapping
E0[3]	PCIe3				*PCIe4*	PCIe1	SG5	SG6	1111 1111	36	SRIO2				Aurora		SATA1	SATA2	1111 1122
F2[3]	PCIe3	SG10	SG1	SG2	*PCIe4*	PCIe1	PCIe2	SG6	1111 1111	36	SRIO2				Aurora		SATA1	SATA2	1111 1122
F8	PCIe3	SG10	*SG1*	*SG2*	PCIe4	PCIe1	PCIe2	SG6	1122 1111	36	SRIO2				Aurora		SATA1	SATA2	1111 1122
FA	PCIe3	SG10	*SG1*	*SG2*	PCIe4	PCIe1	SG5	SG6	1122 1111	36	SRIO2				Aurora		SATA1	SATA2	1111 1122
CB	PCIe3	SG10	*SG1*	*SG2*	PCIe4	SG4	SG5	SG6	1122 1111	1F[1]	*PCIe1*				PCIe2				1111 2222
67	XFI9	XFI10	XFI1	XFI2	*PCIe4*				1111 2222	1F[1]	*PCIe1*				PCIe2				1111 2222
AB	PCIe3				*PCIe4*				1111 1111	02	PCIe1								1111 1111
AB	PCIe3				*PCIe4*				1111 1111	29	*SRIO2*				*SRIO1*				1111 1111
DB	PCIe3								1111 1111	1F[1]	*PCIe1*				PCIe2				1111 2222
DB	PCIe3								1111 1111	2D[1]	SRIO2				SRIO1				1111 1111
66	XFI9	XFI10	XFI1	XFI2	*PCIe4*				1111 2222	16	PCIe1				PCIe2		SATA1	SATA2	1111 1122
94	SG9	SG10	*SG1*	*SG2*	*SG3*	SG4	SG5	SG6	1122 1111	27	*PCIe1*						SATA1	SATA2	1111 xx22
66	XFI9	XFI10	XFI1	XFI2	*PCIe4*				1111 2222	15	PCIe1				PCIe2		SATA1	SATA2	1111 1122
66	XFI9	XFI10	XFI1	XFI2	*PCIe4*				1111 2222	27	*PCIe1*						SATA1	SATA2	1111 xx22
66	XFI9	XFI10	XFI1	XFI2	*PCIe4*				1111 2222	1F	*PCIe1*				PCIe2				1111 2222
AA[3]	PCIe3				*PCIe4*				1111 1111	1F[1]	*PCIe1*				PCIe2				1111 2222
6B	XFI9	XFI10	XFI1	XFI2	PCIe4	SG4	SG5	SG6	1111 2222	16	PCIe1				PCIe2		SATA1	SATA2	1111 1122

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Table 19-1. SerDes Lanes Assignments and Multiplexing (continued)

SERDES1										SERDES2									
SRDS_PRTCL_S1	A	B	C	D	E	F	G	H	Per Lane PLL Mapping	SRDS_PRTCL_S2	A	B	C	D	E	F	G	H	Per Lane PLL Mapping
6A	XFI9	XFI10	XFI1	XFI2	PCIe4	SG4	SG5	SG6	1111 2222	15	PCIe1				PCIe2		SATA1	SATA2	1111 1122
6A	XFI9	XFI10	XFI1	XFI2	PCIe4	SG4	SG5	SG6	1111 2222	1F[1]	*PCIe1*				PCIe2				1111 2222
D9	PCIe3	SG10	SG1	SG2	PCIe4	SG4	SG5	SG6	1111 2111	36	SRIO2				Aurora		SATA1	SATA2	1111 1122
51	XAUI9				PCIe4	SG4	SG5	SG6	1111 2222	36	SRIO2				Aurora		SATA1	SATA2	1111 1122
66	XFI9	XFI10	XFI1	XFI2	*PCIe4*				1111 2222	01	PCIe1								1111 1111
A2	SG9	SG10	SG1	SG2	SG3	SG4	SG5	SG6	1111 1111	27	*PCIe1*						SATA1	SATA2	1111 xx22
DA	PCIe3								1111 1111	16	PCIe1				PCIe2		SATA1	SATA2	1111 1122

1. Using PBI command, set SRDS2PLL2CR1=0x800_4100.
2. Using PBI command, set SRDS1PLL1CR1=0x800_4100. The PLL1 can not be disabled. The RCW field SRDS_PLL_REF_CLK_SEL_S1 must be configured to 100 MHz or 125 MHz to apply the reference clock to both PLL 1 and PLL2.
3. For options with Gen 3 PCIe, PLL2 is automatically enabled during speed switch. Therefore, the system must supply a reference clock to PLL2. The RCW field SRDS_PLL_REF_CLK_SEL_S1 must be configured to 100 MHz or 125 MHz to apply the reference clock to PLL2.

19.1.1.1.2 Rules for Protocol Selection

- RGMII may use MAC3 and/or MAC4/10. When RGMII is enabled for a particular MAC (using EC1/EC2 RCW fields), it always takes precedence over SerDes-based interfaces.
- If SRIO2 is chosen in SRDS_PRTCL_S2 then DMA3 is not functional. If SRIO2 is not chosen DMA3 is functional.
- Only PCIe1 supports SR-IOV.
- Only one PCIe in the device can support 8/5/2.5 GT/s. SRDS1 (PCIe4) and SRDS2 (PCIe1) may support these frequencies. Any combination in which both PCIe controllers support 8/5/2.5 GT/s is not allowed.
- In case that PCIe1 was chosen in both Serdes, Serdes2 will be used. It is recommended to powerdown(SW) the Serdes1 lane to save power.